MIL-I-7906A(WEP)
15 JUNE 1961
SUPERSEDING
MIL-I-7906(AER)
21 JULY 1952

MILITARY SPECIFICATION

INDICATOR, GYRO HORIZON, MANUALLY CAGED, ELECTRIC, TYPE MB-1

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers certain design and all performance requirements for electric, manually caged, gyro horizon indicators.

1.2 Classification - This specification covers one type of gyro horizon indicator designated as Type MB-1.

2. APPLICABLE DOCUMENTS

2.1 General - The following specifications, standards and publications of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-N-290	Nickel Plating (Electrodeposited)
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
DD-G-451	Glass, Flat and Corrugated for Glazing, Mirrors and other Uses

Military

MIL-P-116	Preservation, Method of
MIL-N-3336	Nut, Self Locking, Instrument Mounting
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-I-6181	Interference, Control Requirements, Aircraft Equipment

FSC 6610

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MIL-L-6880	Lubrication of Aircraft, General Specification for
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS 33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS 33586	Metals, Definition of Dissimilar
MS 17313	Indicator-Gyro Horizon, Electric, Type MB-1

2.2 Availability of Documents

(1) When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Qualification - The gyro horizon indicator furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed on, or approved for listing on, the applicable qualified products list.

3.2 Selection of Government Documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial Parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

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b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard Parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical Materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Metals - Metals shall be of the corrosion-resistant type, or shall be suitable protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the instrument may be subject when in storage or during normal service life.

3.3.2.1 Dissimilar Metals - Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MS 33586 shall not be used in intimate contact with each other.

3.3.3 Nonmagnetic Materials - Nonmagnetic materials shall be used for all parts of the instrument except where magnetic materials are essential.

3.3.4 Fungus-Proof Materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed they shall be treated with a fungicidal agent acceptable to the Bureau of Naval Weapons.

3.4 Design and Construction - The indicator shall be designed to indicate the attitude of an airplane throughout all maneuvers and to permit loops and rolls to be made without tumbling the gyro. The indicator shall incorporate a device for sensitive indications in pitch and a means to prevent fogging on the surfaces of the cover glass under extremes of atmospheric conditions. The means provided shall be subject to approval by the Bureau of Naval Weapons. The design shall be reasonably simple, to permit overhaul or repair without requiring numerous special tools and fixtures. The indicator shall be constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation and services.

3.4.1 Case - The indicator case shall conform to MS 17313 . It shall be made of aluminum alloy uniform in texture, have a smooth surface and shall be suitably protected against the infiltration of dust and sand particles. The bezel ring on the front of the case shall be held in place by means of screws and by its removal the cover glass shall be replaceable. The case shall be suitable for front or rear mounting on an instrument panel. The case shall be finished with a durable lusterless black finish, Color No. 37038 of FED-STD-595.

3.4.2 Gyro Parts - The gyro parts shall be mounted so that the instrument cover may be removed without disturbing the adjustments.

3.4.3 Erection Device - The instrument shall incorporate an erection device to maintain the gyro in its neutral position when running.

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3.4.4 Caging Device - A caging device shall be incorporated into the indicator and shall be an integral part thereof. The mechanism shall be constructed so that the gyro may be caged manually at any time from any indication and returned to straight and level flight indication by means of mechanical levers. This mechanism shall be self-retracting.

3.4.5 Pitch Indications - The instrument shall incorporate two separate indications in pitch, a horizon bar and a pitch scale. The horizon bar shall indicate the pitch attitude to not less than 25 degrees in climb and dive and the pitch scale shall indicate the pitch attitude within a range of from 10 to 80 degrees in climb and dive.

3.4.5.1 Zero Pitch Adjustment - The indications of zero pitch shall be adjusted by means of an external knob within a minimum range of from 6 degrees nose down to 20 degrees nose up from the normal position. The normal position with respect to the mounting lugs shall be suitably marked with an index.

3.4.6 Bank Index - The indicator shall incorporate a bank index which shall not be affected by pitch deviations of the gyro.

3.4.7 Turn Error Compensation - The indicator shall be compensated to maintain to a minimum the error in indication due to centrifugal force encountered in turn. Unless otherwise approved by the Bureau of Naval Weapons, the method of compensation shall consist of inclining the top of the spin axis of the gyro toward the direction of flight.

3.4.8 Wiring - The complete electrical wiring system shall be insulated from the indicator case.

3.4.8.1 Electrical Connector - The electrical connector shall be in accordance with Specification MIL-C-5015 except that the connector used may have the mounting flange contour modified if required for mounting considerations.

3.4.8.2 Flag Alarm - A warning device shall be incorporated which will indicate when phase rotation is incorrect or when power failure occurs.

3.4.9 Dial -

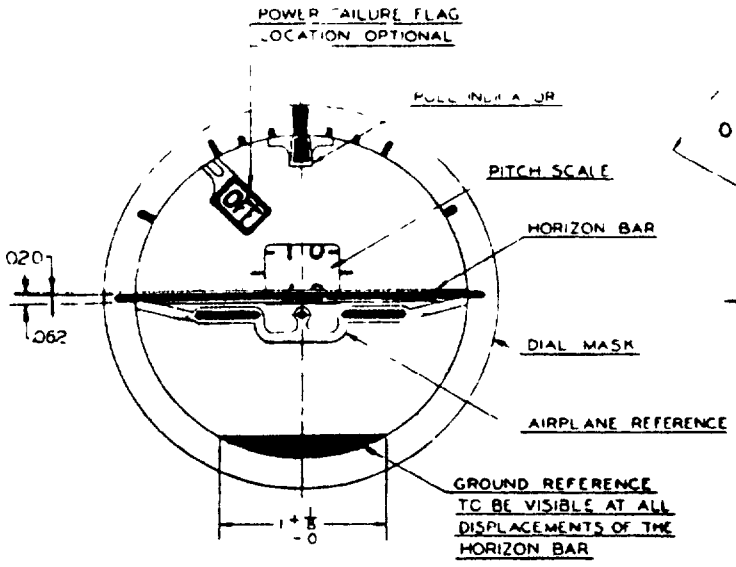
3.4.9.1 Cover Glass - The quality of the cover glass shall be in accordance with Specification DD-G-451, Type II, Quality AA. The cover glass shall be properly sealed to prevent fogging.

3.4.9.2 Visibility of Dial - The horizon bar, bank index, numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

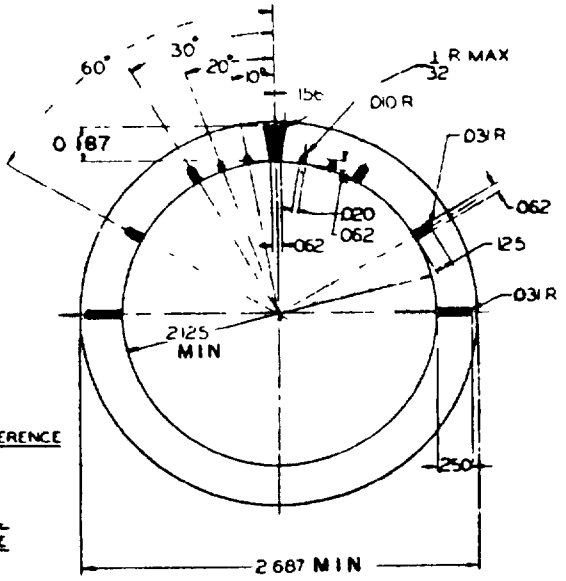
3.4.9.3 Markings - The background of the dial assembly of the indicator shall be finished in a durable, lusterless black material of Color No. 37038 of FED-STD-595. The markings shown in Figure 1 in solid black shall be finished in a durable, lusterless white material of Color No. 37875 of FED-STD-595. The form of the letters and numerals on the dial shall conform to MS 33558 and the dimensions of the markings shall be as specified in Figure 1.

3.4.10 Maintenance - The design shall be such as to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly by those tools and items of maintenance equipment which are normally available as commercial standards.

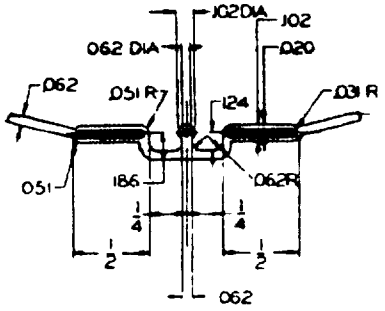
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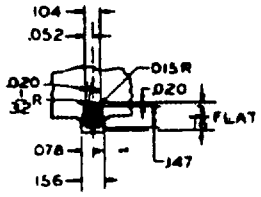
DIAL ASSEMBLY



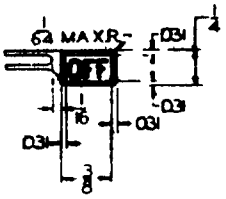
DIAL MASK



AIRPLANE REFERENCE

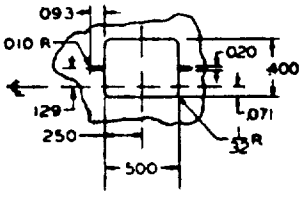


ROLL INDICATOR

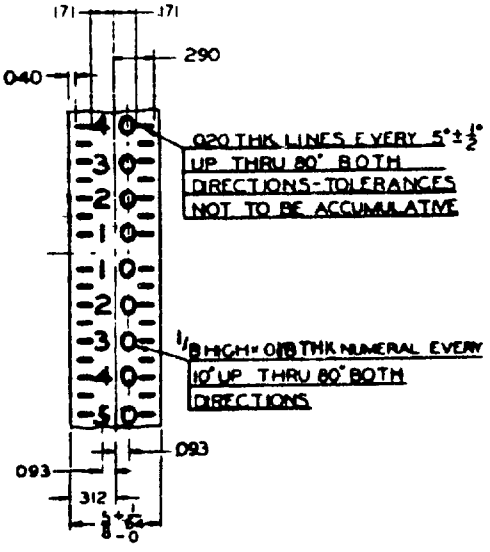


LETTERS 1/8 HIGH × 018 THK LINES

POWER FAILURE FLAG



PITCH SCALE OPENING



PITCH SCALE

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES FRACTIONS ± 1/64, DECIMALS ± .010, ANGLES ± 1°

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3.4.11 Weight - The weight of the indicator shall not exceed 4.5 pounds.

3.5 Performance - The indicator shall perform satisfactorily when subjected to the tests specified in Section 4.

3.6 Physical Requirements -

3.6.1 Finish - Protective coatings and finishes which will crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.6.2 Aluminum Alloy Parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625.

3.6.3 Brass and Steel Parts - Brass and steel parts shall be nickel, chromium or cadmium plated where practicable in accordance with Specifications QQ-N-290, QQ-C-320 and QQ-P-416, respectively.

3.6.4 Lubrication - Lubrication shall be accomplished in accordance with Specification MIL-L-6880 except that the selection of the lubricants used shall be approved by the Bureau of Naval Weapons.

3.6.5 Screw Threads - Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.6.6 Workmanship - The instrument, including all parts and accessories, shall be constructed and finished to produce an instrument free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.7 Identification of the Product -

3.7.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be durably marked to withstand usage encountered in service. The nameplate shall be legibly marked with the following information:

BUWEP, U. S. NAVY
INDICATOR, GYRO HORIZON, TYPE MB-1
MS 17313-1
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract/Order No.
Manufacturer's Name

3.7.2 Manufacturer's Part Number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

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3.7.3 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.8 Installation -

3.8.1 Installation Instruction - The contractor shall furnish with each instrument one printed copy of instructions with illustrations and diagrams, if necessary, covering the installation of the instrument. Previous to printing, two copies shall be furnished to the Bureau of Naval Weapons for approval. Instructions shall be printed on 8-1/2 by 11 inch paper or 17 by 11 inch paper.

3.8.2 Mounting Screws - The contractor shall furnish four mounting screws for installing the instrument. They shall be No. 6-32 NC-2 fillister head nonmagnetic machine screws having durable dull black oxidized or black nickel finish. Length shall be sufficient for mounting on a panel 5/16 inch thick.

3.8.3 Mounting Nuts - The contractor shall furnish four No. 6-32 NC-2 removable spring nuts conforming to Specification MIL-N-3336.

3.8.4 Envelope - An envelope furnished by the contractor containing installation instructions, mounting nuts and mounting screws shall be packaged with each instrument, and shall be marked with the following information.

"IMPORTANT"

"THIS ENVELOPE CONTAINS INSTRUCTIONS,
MOUNTING NUTS, AND MOUNTING SCREWS"

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests - The Inspection and Testing of the gyro horizon indicators shall be classified as follows

a. Qualification Tests - Qualification tests are those tests performed on samples submitted for qualification as a satisfactory product.

b. Acceptance Tests - Acceptance tests are those tests performed on indicators manufactured and submitted for acceptance under contract.

4.3 Qualification Tests - The qualification tests of the indicators shall consist of all the tests of this specification specified under the paragraph headed Test Methods.

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4.3.1 Qualification Test Instructions - Qualification test samples shall consist of three indicators of each manufacturer's part number to be furnished in accordance with this specification. The instruments submitted for testing shall not have been previously tested except for the Individual Tests. The samples shall be forwarded to the testing laboratory designated in the letter of authorization

4.3.2 Test Sample Identification - The test samples shall be plainly identified by securely attached, durable tags marked with the information listed below.

Sample for Qualification Testing
Indicator, Gyro Horizon, Manually Caged
Submitted by (Manufacturer's Name, date) for Qualification
Testing in accordance with Specification MIL-I-7906A
under authorization (reference letter authorizing tests).
MS 17313-1
Type Number MB-1
Manufacturer's Part Number

4.4 Acceptance Tests - The acceptance tests of the indicator shall consist of the Individual Tests and the Sampling Plans of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual Tests - The individual tests shall consist of the tests listed below, conducted in the order given, on each indicator.

Inspection
Current Consumption
Alignment
Erection Rate
Roll, Pitch and Yaw
Axis Inclination
Low Voltage and Power Failure Operation
Loop and Roll
Dielectric Strength

4.4.2 Sampling Plans and Tests -

4.4.2.1 Sampling Plan A - Samples shall consist of three gyro horizon indicators selected at random by the Inspector from each lot of 100 or less which have passed the Individual Tests. A lot shall consist of identical instruments with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Instruments which have been subjected to the Sampling Plan A tests shall not be delivered on contract until they have been reconditioned and resubmitted to all Individual Tests.

4.4.2.1.1 Sampling Plan A Tests - The Sampling Plan A tests shall consist of the tests listed below conducted in the order given on each sample.

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Horizon Bar sensitivity
Pitch Scale Error
Magnetic Effect
Vibration Error
Low Temperature Operation
High Temperature Operation

4.4.2.2 Sampling Plan B - Unless otherwise specified, the samples shall consist of three gyro horizon indicators selected at random by the Inspector from the first fifteen instruments produced on contract within ten days after manufacture. The samples shall be forwarded to the laboratory specified herein (See 6.2) and shall be plainly identified by a securely attached, durable tag marked with the following information

Indicator, Gyro Horizon, Manually Caged, Type MB-1
Submitted by (manufacturer's name, date) for production acceptance test in accordance with Contract No. _____.

4.4.2.2.1 Sampling Plan B Tests - The sampling plan B tests shall consist of the following tests conducted in the order given on each sample

Individual Tests
Sampling Plan A Tests
Radio Interference
High Altitude-Low Temperature
Humidity

4.4.3 Rejection and Retest Provisions -

4.4.3.1 Individual Test Failure - Any indicator failing to meet the requirements of the Individual Tests shall be rejected and returned at the contractor's expense. Instruments which have been rejected may be reworked or replaced to correct the deficiencies and resubmitted for all specified tests. Before resubmissions, a report giving full particulars concerning previous rejection and the action taken to correct the defects shall be furnished the Government Inspector. Instruments rejected after retest shall not be resubmitted without the specific approval of the Bureau of Naval Weapons.

4.4.3.2 Sampling Plan Test Failure - When any indicator fails to meet the requirements of any of the tests of the sampling plans, the lot represented shall be rejected and returned at the contractor's expense. Acceptance of the lot will be withheld until the extent and cause of failure are determined. Lots rejected after retest shall not be resubmitted without the specific approval of the Bureau of Naval Weapons.

4.4.3.2.1 Individual Test Continuance - For production and operational reasons, the Individual Tests may be continued pending the investigation of a Sampling Plan test failure but final acceptance of instruments on hand or later produced shall not be made until it is determined that the instruments meet all the requirements of this specification.

4.4.3.2.2 Width of Failure Effect - If investigation of the cause of failure of a sampling plan sample indicates that the defect may exist in instruments previously accepted, full particulars concerning the deficiency including recommendations for correction, shall be furnished for the approval of the Bureau of Naval Weapons. Upon approval of the correction a test shall be added to the Individual Tests to check all indicators for this deficiency until reasonable assurance can be obtained that the defect has been satisfactorily corrected.

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4.5 Test Conditions -

4.5.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of approximately 25°C (77°F) When tests are made with atmospheric pressure and temperature substantially different from the above values, proper allowance shall be made for the difference from specified conditions.

4.5.2 Vibration Stand - Whenever a vibration stand is specified, it shall consist of a device which will vibrate at any desired frequency between 300 and 3000 cycles per minute and shall subject the instrument to vibration such that a point on the vibration stand shall describe in a plane inclined 45 degrees to the horizontal plane a circle of the diameter specified herein.

4.5.3 Scorsby (or equivalent apparatus) Tester -

4.5.3.1 Standard Angle Scorsby - It shall provide 5 to 7 oscillations per minute and subject the instrument to 15 degree roll, pitch and yaw movements. The 15 degree movement shall be considered 7-1/2 degrees on each side of each reference axis.

4.5.3.2 Low Angle Scorsby - It shall provide 5 to 7 oscillations per minute and subject the instrument to 3 degree roll, pitch and yaw movements. The 3 degree movement shall be considered 1-1/2 degrees on each side of each reference axis.

4.5.4 Level Position - Whenever level position is specified, the instrument shall be in a position such that the upper mounting holes are on a horizontal line and face is vertical.

4.5.5 Test Voltage - Unless otherwise specified, all tests shall be conducted with the gyro operating on 115 ± 5 volts, 3 phase, 400 ± 20 cycles alternating current, phase rotation ABC.

4.5.6 Pitch Readings - Unless otherwise specified, the horizon bar shall be used for determining the pitch indications. The bar shall move approximately 0.035 inches per degree of gyro pitch movement at the level position.

4.5.7 Equilibrium - Whenever equilibrium is specified, the test voltage shall be applied and after 15 seconds the instrument shall be caged and released then allowed to operate for 10 minutes.

4.5.8 Caging - When the gyro is being caged, the caging device shall be held in the caged position for sufficient time to permit the horizon bar and bank index to settle down to approximately their zero indications.

4.6 Test Methods -

4.6.1 Examination of Product - Each instrument shall be examined externally to determine conformance with the requirements of this specification with respect to materials, design and construction, physical requirements, marking, workmanship, and installation.

4.6.2 Current Consumption - With the test voltage applied to the terminals and the instrument running not less than 3 minutes, the current shall not exceed 0.125 amperes on the A and B legs and 0.145 amperes on the C leg.

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4.6.3 Alignment - The instrument shall be mounted in a level position and the gyro not running the test voltage shall be applied to the receptacle. After 15 seconds, the gyro shall be caged and immediately released and the indicator shall settle to its zero position within plus or minus one degree in roll and pitch in a time not to exceed 2 minutes.

4.6.3.1 Pitch Scale - The instrument shall be mounted in a level position on a tilt table and operated at equilibrium. The table shall be tilted so that the pitch scale reads 10 degrees climb The angle through which the table was rotated to obtain this reading shall be 10 ± 2 degrees.

4.6.4 Erection Rate -

4.6.4.1 Climbing and Diving - The instrument shall be in the level position and operating at equilibrium. The gyro shall be caged and the instrument displaced slightly beyond 10 degrees in climb. The gyro shall then be uncaged and the instrument returned to level position. The elapsed time required for the gyro to erect from 10 to 5 degree climb indication shall be noted. The test shall be repeated with the gyro displaced to indicate slightly beyond 10 degree dive position and the time required for the gyro to erect from 10 to 5 degree dive indication shall be noted. These elapsed times shall be between 110 and 220 seconds. This test may be performed on a low angle Scorsby.

4.6.4.2 Banking - The instrument shall be in the level position and operating at equilibrium. The gyro shall be caged and the instrument displaced slightly beyond 10 degrees in left bank. The gyro shall then be uncaged and the instrument returned to level position. The elapsed time required for the gyro to errect from 10 to 5 degree left bank indication shall be noted. The test shall be repeated with the gyro displaced to indicate slightly beyond 10 degree right bank and the time required for the gyro from 10 to a 5 degree right bank indication shall be noted. These elapsed times shall be between 110 and 220 seconds. This test may be performed on a low angle Scorsby.

4.6.4.3 Spiral - The instrument shall be in the level position and operating at equilibrium. The gyro shall be caged and the instrument displaced 50° in right bank. The gyro shall then be uncaged and the instrument returned to level position. The gyro shall then be permitted to erect for 2 minutes and the pitch precession noted. The precession in pitch shall not exceed a rate of 1-1/2° per minute. This test shall be repeated for 40° and 30° right bank, and for 50°, 40° and 30° left bank.

4.6.5 Roll, Pitch and Yaw - The indicator shall be mounted level on a standard angle Scorsby apparatus and shall be operated at equilibrium. The indicator shall then be subjected to a 15 degree roll, pitch and yaw movement for 10 minutes. At the end of this period, the Scorsby apparatus shall be leveled and the indicator shall indicate within plus or minus one degree of its zero position in roll and pitch.

4.6.6 Axis Inclination - In instruments incorporating an inclined axis, the instrument shall be mounted on an accurately leveled turnable and operated at equilibrium and the pitch indication shall be noted. The instrument shall be quickly rotated 180 degrees in azimuth and the pitch indicator again noted. The difference between the two indicators shall be twice the angle of axis inclination ±1°.

4.6.7 Low Voltage and Power Failure Operation - A potential of 90 ± 5 volts alternating current shall be applied to the receptacle with gyro not running, after 15 seconds, the gyro shall be caged and released. The indicator shall settle to its zero position within plus or minus one degree in roll and pitch within a period of 3 minutes. The power failure flag shall indicate "off" when the phase rotation is incorrect, and/or when any or all phases are broken.

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4.6.8 Loop and Roll - The indicator shall be mounted in a level position on a test fixture permitting rotation about the horizontal and lateral axis and allowed to reach equilibrium. The indicator shall be looped through one complete revolution (inside loop) at a rate of 15 to 20 degrees per second. The indicator shall indicate zero roll and pitch within 5 degrees immediately after this maneuver. The indicator shall then be rolled through one complete revolution at a rate of 10 ±1 degree per second. The indicator shall indicate zero roll and pitch within one degree immediately after this maneuver.

4.6.9 Dielectric Strength - The indicator shall be subjected to a dielectric test with 500 volts rms alternating current at commercial frequency applied between each terminal and the indicator case for a period of 5 seconds. No breakdown shall occur as a result of this test.

4.6.10 Horizon Bar Sensitivity - The indicator shall be mounted in a level position on a tilt table and allowed to reach equilibrium. This displacement of the horizon bar for angles of climb shall be downward and for angles of dive shall be upward. The horizon bar shall be visible at the maximum displacement of the indicator. The displacement of the horizon bar from the zero position shall be as specified in Table I.

TABLE I

HORIZON BAR SENSITIVITY

Pitch Angle (Degrees)	Displacement (Inches)	Tolerance (Inches)
5	0.175	± 0.050
10	0.340	± 0.050
15	0.495	± 0.050
20	0.625	± 0.050
25	0.740	± 0.050

4.6.11 Pitch Scale Error - The instrument shall be mounted in a level position on a tilt table and operated at equilibrium. The table shall be tilted so that the pitch scale reads 10, 20, 30, 40, 50, 60, 70, 80 degrees in climb and then in dive. The angles through which the table has been tilted shall agree with the respective readings of the pitch scale within plus or minus 2 degrees.

4.6.12 Magnetic Effect - The indicator shall be held in various positions with its nearest part 5-1/2 inches from the center of a short bar-magnet compass whose compensator magnets have been removed and which shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The various positions shall be so chosen as to include that position which causes the maximum deflection of the compass. This test shall be made first with no voltage applied to the indicator and then shall be repeated with the indicator operating at equilibrium. The maximum deflection shall not exceed 5 degrees.

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4.6.13 Vibration Error - The indicator shall be mounted on a vibration stand in its level position. The instrument shall be operated at equilibrium and the roll and pitch indications noted. The instrument shall be subjected to vibration as specified herein with a circle diameter of between 0.003 and 0.005 of an inch at frequencies from 300 to 3000 cycles per minute. The deviations of neither the roll nor pitch indication shall exceed one degree from the indication with no vibration at any frequency within the specified frequency range.

4.6.14 Low Temperature Operation - The indicator not operating shall be subjected to an ambient temperature of -55°C (-67°F) for a period of 4 hours. At the end of this exposure period and with the instrument maintained at -55°C (-67°F) the test voltage shall be applied to the indicator. After 15 seconds the indicator-shall be caged and released and shall settle to zero ± 3 degrees in pitch, and zero ± 3 degrees in roll within 3 minutes and shall remain with this tolerance for 10 minutes, and 15 minutes after caging the indicator shall settle to zero ±1 degree in pitch and zero ±1 degree in roll. This test may be performed on a low angle Scorsby.

4.6.15 High Temperature Operation - The procedure specified for the Low Temperature Operation test shall be repeated except the ambient temperature shall be 70°C (158°F) and the same tolerances shall apply.

4.6.16 Radio Interference - The indicator shall be operated on test voltage specified herein while mounted on a Scorsby apparatus and subjected to 15 degree roll, pitch and yaw movements. The indicator shall then be tested for conducted and radiated radio noise interference in accordance with Specification MIL-I-6181.

4.6.17 Flight Test -

4.6.17.1 Turn Error - The indicator shall be subjected to a flight test consisting of 180 degree turns at a rate of 180 degrees per minute with a true airspeed of between 50 and 500 knots. The error immediately at the completion of the turn shall not exceed 2 degrees in pitch and/or roll. The instrument shall indicate the vertical within one degree of pitch and roll in not more than 3 minutes after the completion of the turn.

4.6.17.2 Loop and Roll - The instrument shall be subjected to a flight test consisting of inside loops and right and left rolls. The error in pitch and/or roll immediately after each loop, shall not exceed 10 degrees and the instrument shall indicate the vertical within one degree of pitch and roll in not more than 5 minutes after the completion of the maneuver. The error in pitch and/or roll immediately after each roll shall not exceed 2 degrees and the instrument shall indicate the vertical within one degree of pitch and roll in not more than one minute after the completion of the maneuver. The lateral axis of the airplane shall remain horizontal within 2 degrees throughout loops. The loops shall be made at a rate of 10 degrees per second or less.

4.6.18 High Altitude - Low Temperature - The indicator not operating, mounted in a level position, shall be placed in a chamber maintained at a temperature of -55°C (-67°F) and subjected to a pressure altitude condition of 65,000 feet (1.68 inches of mercury absolute pressure) for a period of 4 hours. At the end of this period, while still 65,000 feet, the indicator shall be subjected to the "Low Temperature Operation" test and meet the requirements specified therein.

4.6.19 Low Temperature Exposure - The indicator not operating, mounted in a level position, shall be subjected to a temperature of -65 ± 2°C (-85 ± 3.6°F) for a period of 48 hours. The temperature shall then be raised to -55 ± 2°C (-67 ± 3.6°F) for a period of 24 hours. The equipment shall then be subjected to the "Low Temperature Operation" test. After the chamber has been returned to room temperature and the indicator has saturated at room

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temperature for a minimum period of 4 hours, it shall be subject to and shall meet the requirements of the "Roll, Pitch and Yaw" test. There shall be no damage which would adversely affect subsequent operation. This test may be combined with the "Low Temperature Operation" test at the direction of the testing agency.

4.6.20 High Temperature Exposure - The indicator not operating, mounted in a level position, shall be subjected to a temperature of 70 ± 2°C (158 ± 3.6°F) for a period of 24 hours. At the end of the 24 hours and while still at the high temperature, the indicator shall then be subjected to the "High Temperature Operation" test. After the chamber has been returned to room temperature and the indicator has saturated at room temperature for a minimum period of 4 hours, it shall be subjected to and meet the requirements specified in the "Roll, Pitch and Yaw" test. There shall be no damage which would adversely affect subsequent operation. This test may be combined with the "High Temperature Operation" test at the direction of the testing agency.

4.6.21 Vibration Failure - The indicator shall be mounted on a vibration stand in its level operating position. The indicator shall be operated at equilibrium, and shall be subjected to vibration as specified herein with a circle diameter between 0.018 and 0.020 of an inch. The frequency shall be varied at a uniform rate of approximately 90 cycles per minute per minute from 300 to 3000 cycles per minute and 3000 to 300 cycles per minute once each hour during this test. This cycle shall be repeated a sufficient number of times to extend the total time to approximately 3 hours (3 cycles). At the end of this period, the instrument shall be examined and there shall be no evidence of looseness in the mechanism as a result of the vibration. The instrument shall be subjected to and meet the requirements of the Individual Tests.

4.6.22 Humidity - The Erection Rate test at room temperature shall be conducted on the indicator. The indicator, with external connections to simulate installed conditions, shall be mounted in its normal operating position in a chamber and maintained at a temperature of 70 ± 2°C (158. ± 3.6°F) and a relative humidity of 95 ± 5 percent for a period of 6 hours. Distilled water shall be used to obtain the required humidity. At the end of the 6-hour period, the heat shall be shut off and the indicator shall be allowed to cool for a period of 18 hours in this atmosphere, in which the humidity rises to 100 percent as the temperature decreases. This cycle shall be repeated a sufficient number of times to extend the total time of test to 120 hours (5 cycles). At the end of the 120-hour (5 cycle) period and within one hour after the indicator has been removed from the chamber, it shall be subjected to and meet the requirements of the Erection Rate test at room temperature. Examination for corrosion and other deterioration shall be made between 24 and 48 hours after completion of the humidity exposure. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the indicator.

4.6.23 Fungus Resistance - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272, except that the test period shall be 14 days. At the option of the qualifying agency, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled instrument.

4.6.24 Salt Spray - The indicator, with external connections made to simulate installed conditions, shall be subjected to a salt spray in accordance with Specification MIL-E-5272 for a period of 50 hours. At the end of the 50 hours period, the indicator shall be subjected to and shall meet the requirements specified in the Individual Tests.

4.6.25 Mounting Lugs - The indicator case shall be mounted face downward on the removable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for one minute to each lug in a direction toward the front of the case. The lugs shall withstand the applied load without fracture.

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5. PREPARATION FOR DELIVERY

5.1 Packaging -

5.1.1 Level A - The equipment shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method IId, omitting preservation compound and using metal reusable containers (see 6.3).

5.1.2 Level C - When this level is required, packaging shall be in accordance with standard commercial practice

5.2 Packing -

5.2.1 Levels A and B - The instrument shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C - The instrument shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers' rules and regulations in effect at the time of shipment.

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Precautionary Marking - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE

DELICATE INSTRUMENTS

HANDLE WITH CARE

6. NOTES

6.1 Intended Use - The manually caged electric gyro horizon indicators covered by this specification are intended for use in aircraft to indicate the attitude during all maneuvers of the airplane in which this instrument is mounted.

6.2 Drawing Requirements - The attention of contractors in invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

6.3 Ordering Data - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. The quantity and MS part number of the instrument desired.

c. Whether Sampling Plan B is to be omitted and if not, add the name and address of the laboratory to which the samples should be submitted.

d. The levels of packaging (see 5.1) and packing (see 5.2) desired.

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6.4 Qualification With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington 25, D C., and information pertaining to qualification of products may be obtained from that activity.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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